The chemistry of life

Introduction to biomolecules

The chemistry of life: introduction to biomolecules

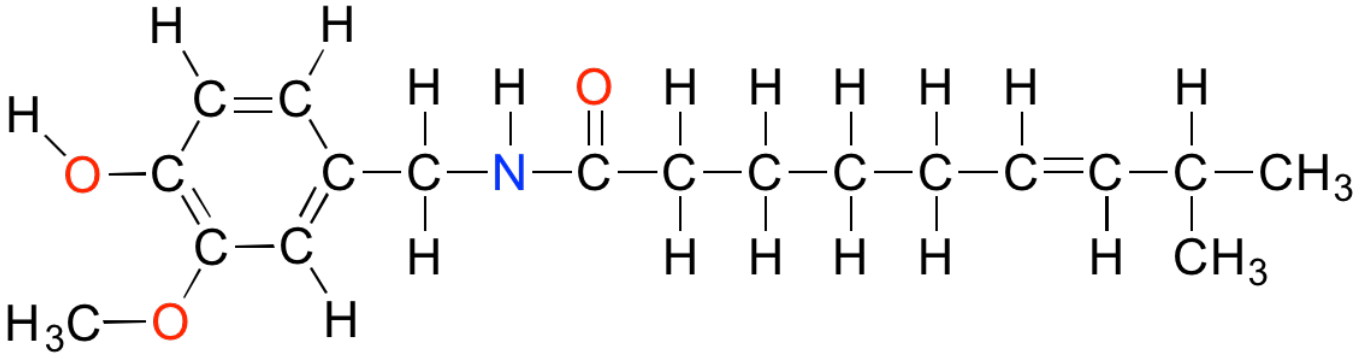
Learning goals:

- What is an organic compound?
- Basic principles of carbon chemistry
- The four groups of organic compounds in biology
- Main functional groups in biomolecules and their nomenclature
- Carbon oxidation states
- Concepts of:
  - Polarity
  - Isomerism
  - Configuration vs. conformation
  - Condensation/Hydrolysis
  - Acids and bases (TBL 2.1.)

Abundance of elements

Human body	%	Earth's crust	%
H	63	O	47
O	25.5	Si	28
C	9.5	Al	7.9
N	1.4	Fe	4.5
Ca	0.31	Ca	3.5
P	0.22	Na	2.5
Cl	0.08	K	2.5
K	0.06	Mg	2.2

The chemistry of life is based on carbon



The four groups of organic compounds found in living organisms are:

- Carbohydrates
- Proteins
- Lipids
- Nucleic acids

But, our body is not only made up of organic compounds:

- H_2O O_2 CO_2 NH_4^+ Ca^{2+} HPO_4^{2-} HCO_3^- ...

Elements essential for life

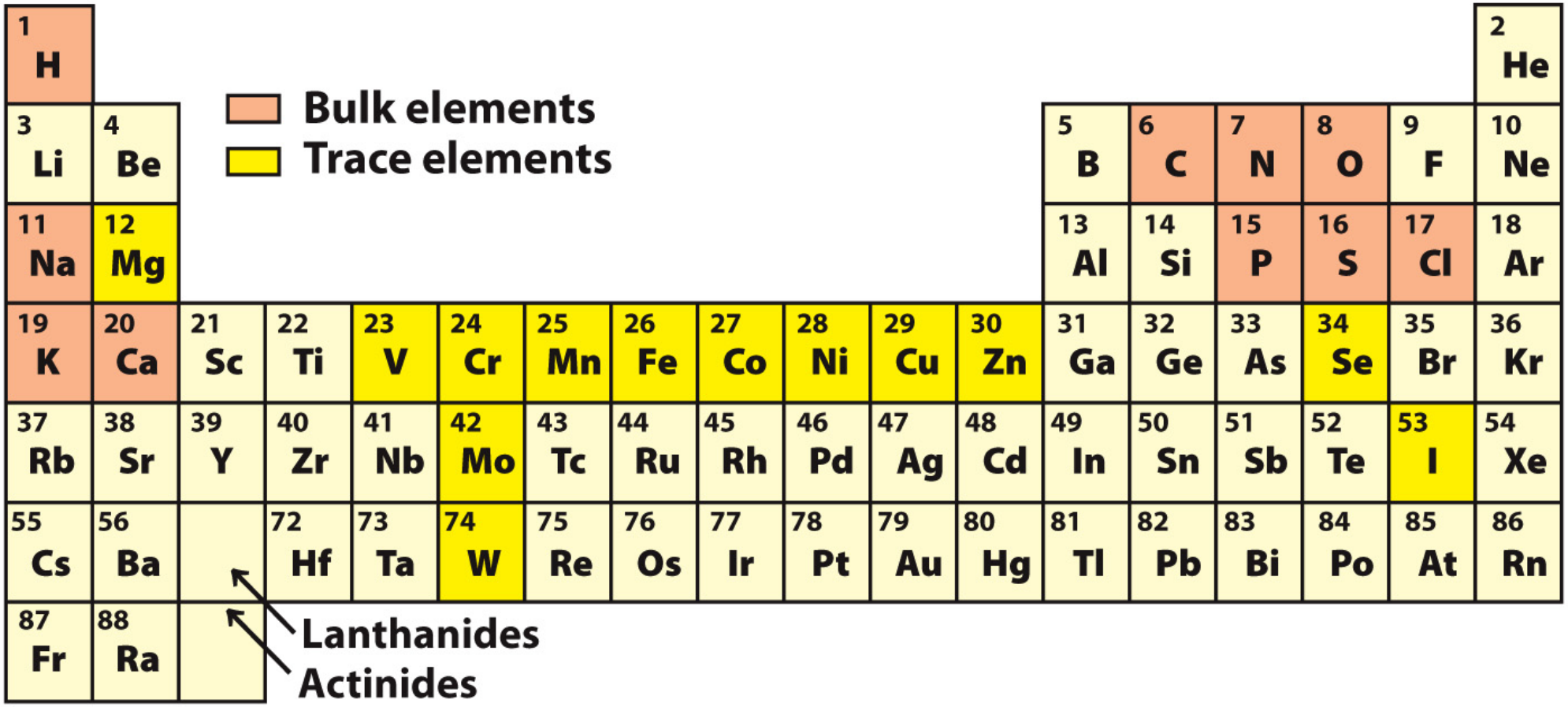
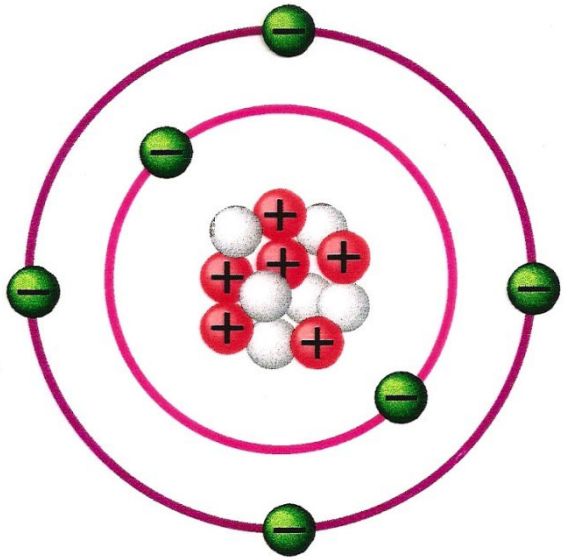


Figure 1-14

Lehninger Principles of Biochemistry, Seventh Edition

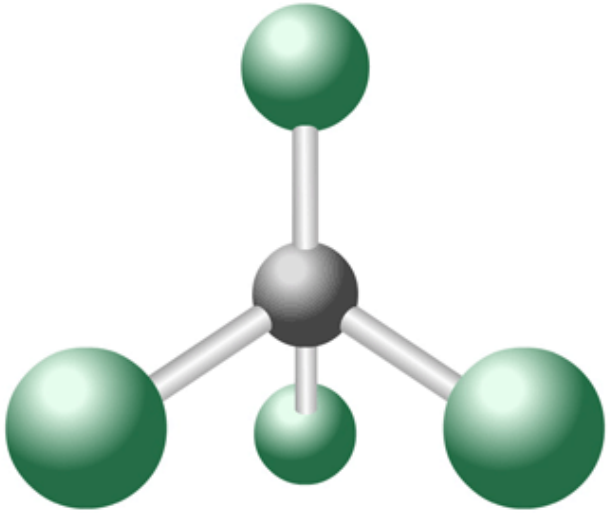
© 2017 W. H. Freeman and Company

Carbon atoms have four electrons to share

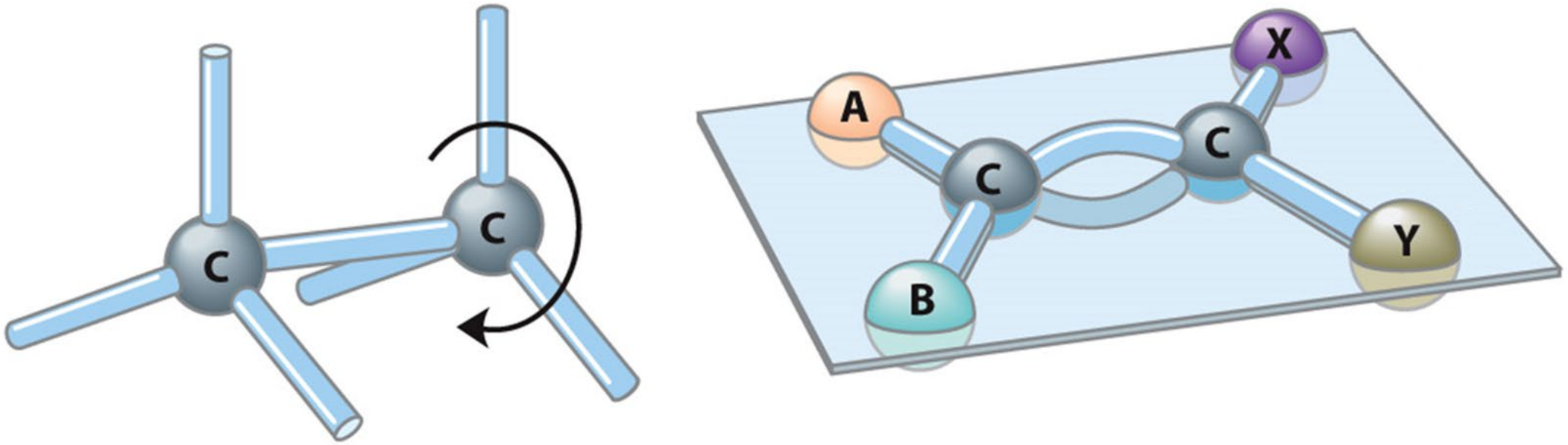


$1s^2\ 2s^2\ 2p^2$

- Carbon atoms have four electrons in their outer electron shell, allowing them to form four covalent bonds with other atoms.
- Covalent bonds are the strongest type of bond between atoms. Stable molecules can be formed.
- Carbon atoms have a characteristic tetrahedral arrangement of the four single covalent bonds (hybrid orbitals sp3 with angles of 109.5º)



Geometry of carbon bonding



- Carbon-carbon single bonds allow free rotation
- Double bounds are shorter and do not allow free rotation

Figure 1-16
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

Versatility of carbon bonding

- Carbon can form covalent carbon-carbon single, double, and triple bonds (rarely in biology)
- Carbon atoms can readily bind to other elements (H, O, N, S, etc.)
- Linked carbon atoms can form linear chains, branched chains and cyclic structures
- Carbon-carbon single bonds allow free rotation

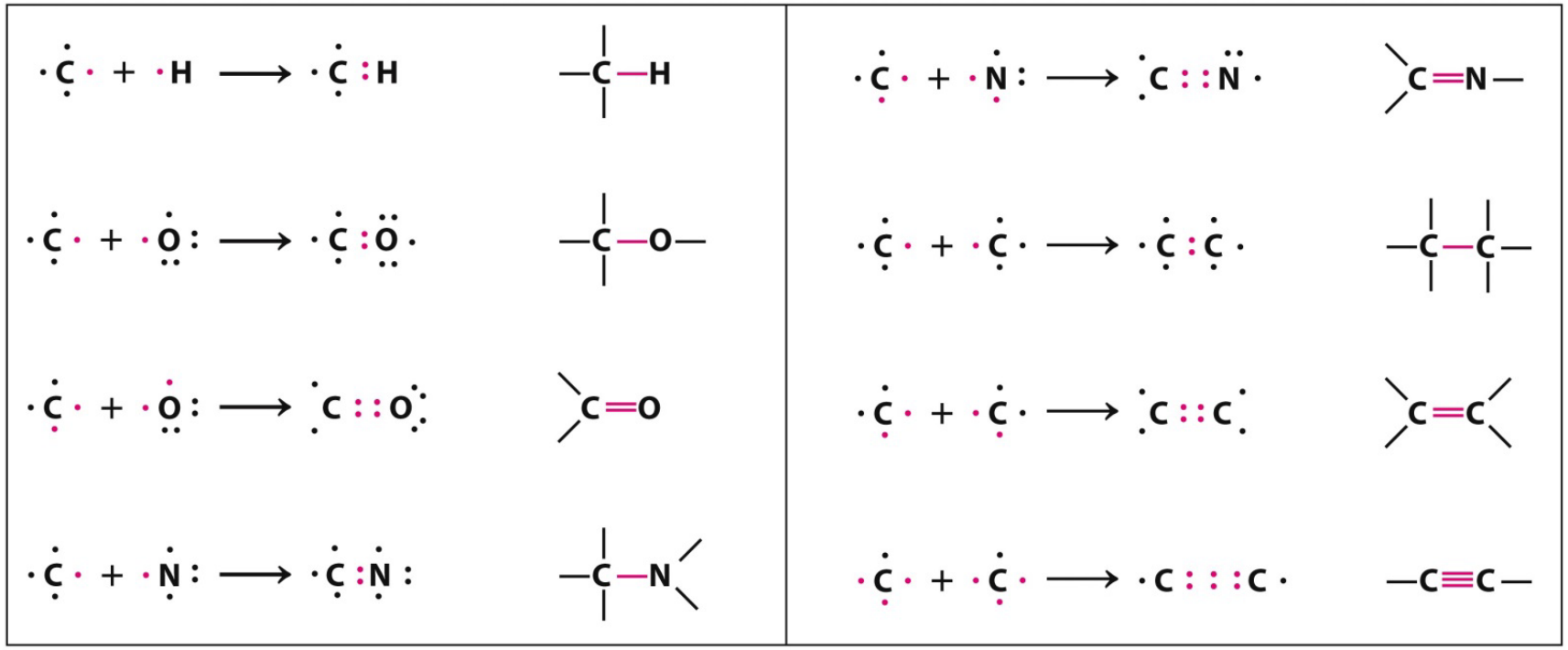


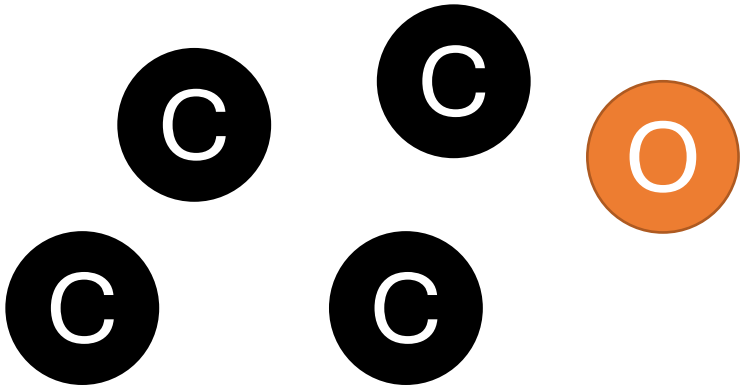
Figure 1-14
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

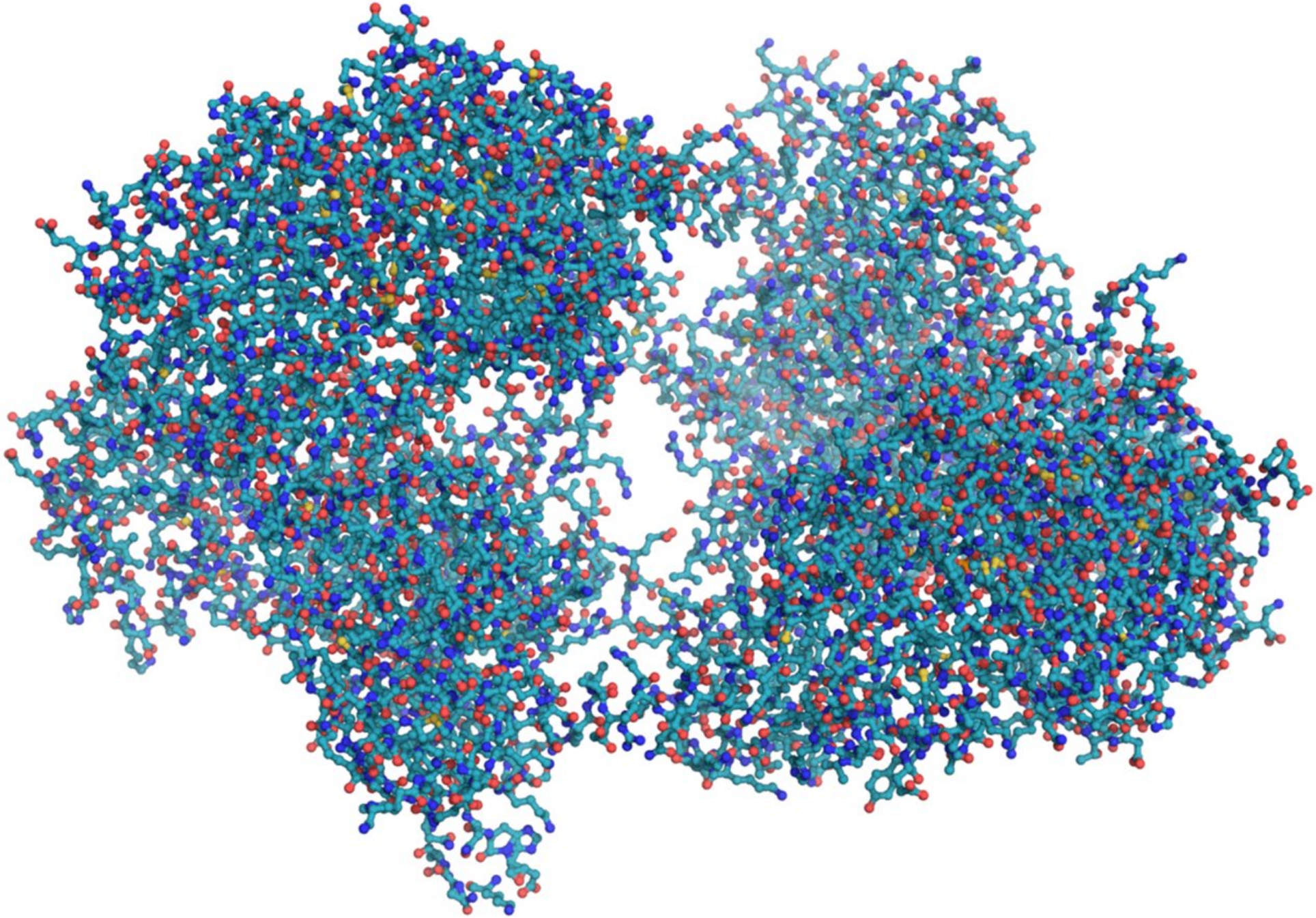
Versatility of carbon bonding

Carbon versatility results in extraordinary structural diversity



How many molecules can be built using 4 carbon atoms (single bonded) and 1 oxygen atom plus the required hydrogen atoms?





Biomolecules typically have several functional groups

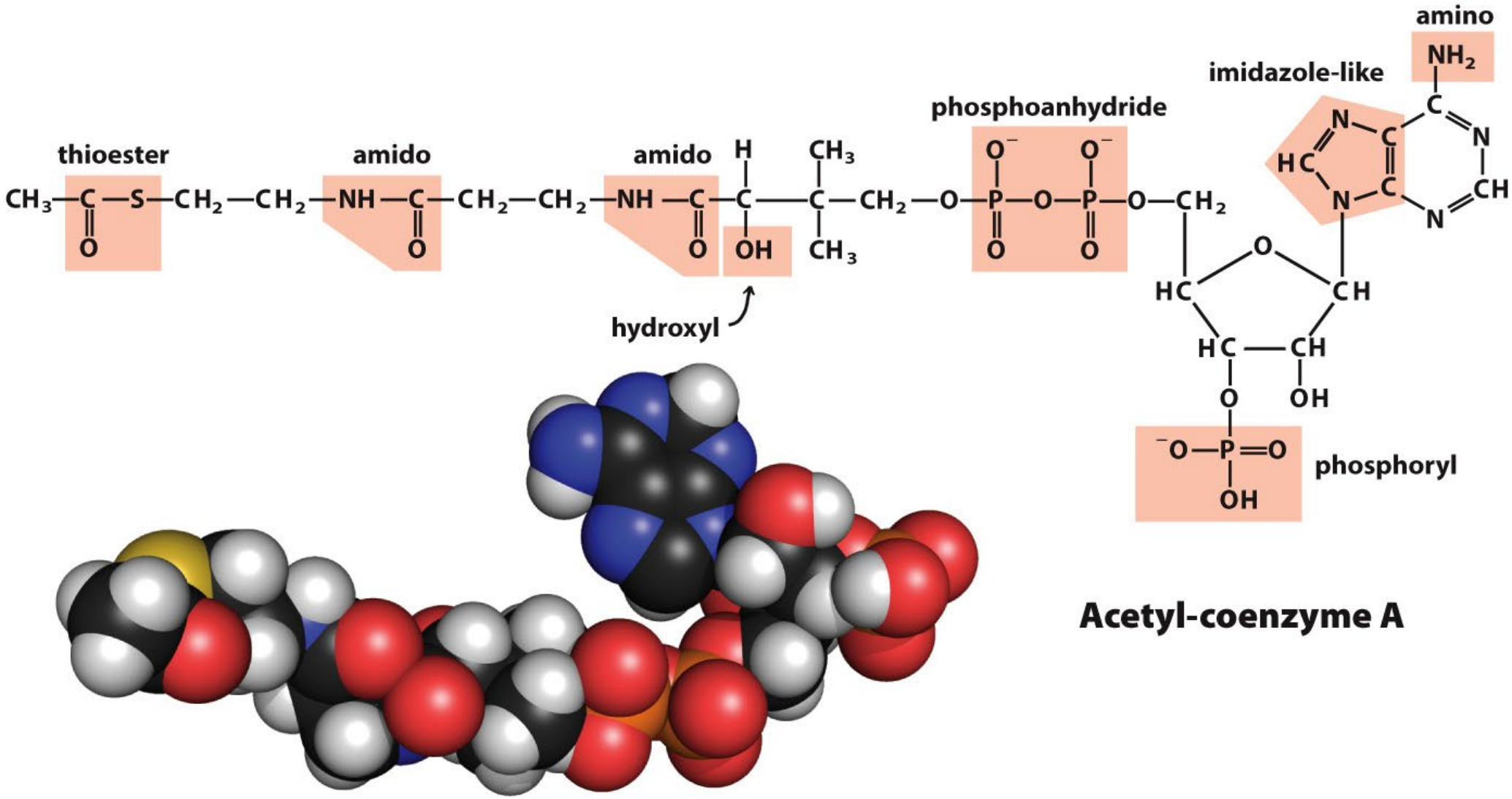
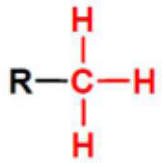
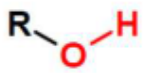


Figure 1-18
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

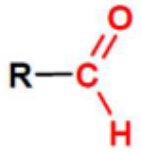
Common functional groups of biomolecules



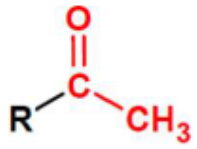
Methyl



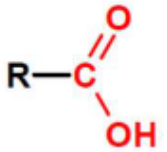
Hydroxyl (alcohol)



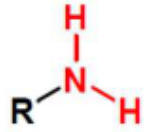
Carbonyl (aldehyde)



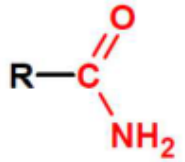
Carbonyl (ketone)



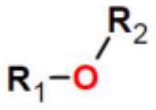
Carboxyl (carboxilyc acid)



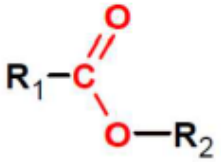
Amine



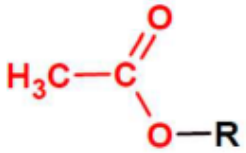
Amide



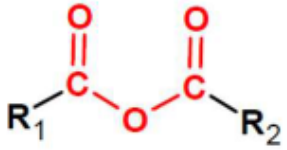
Ether



Ester



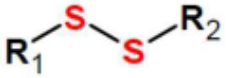
Acetyl



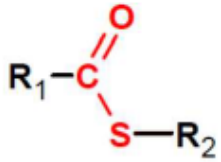
Anhydride



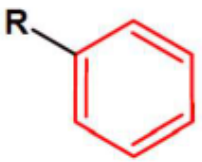
Sulfydryl (thiol)



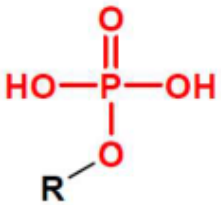
Disulfate



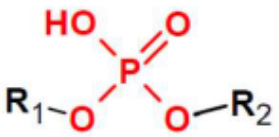
Thioester



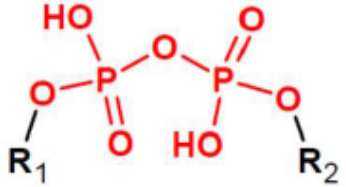
Phenyl



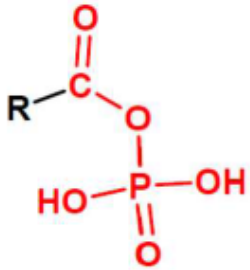
Phosphoester



Phosphodiester



Phosphoanhydride



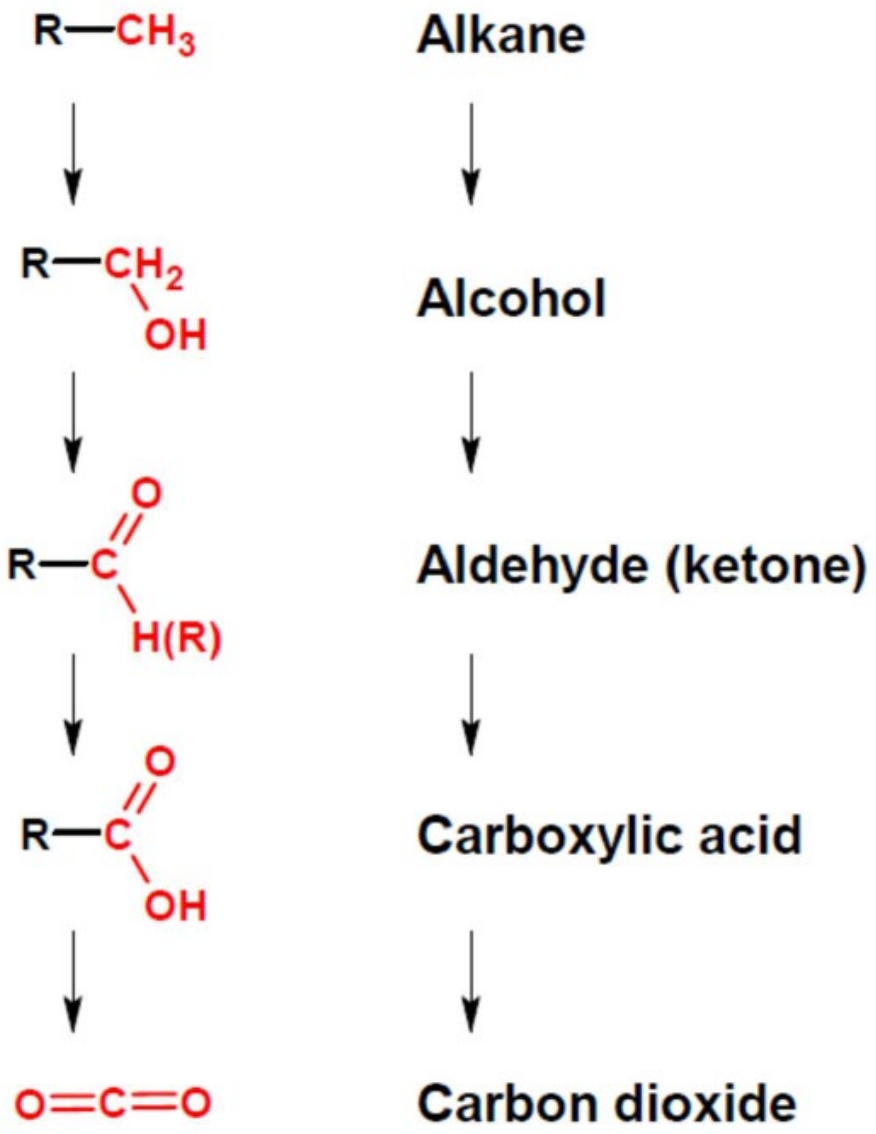
Acyl phosphate

Nomenclature of organic molecules

Prefix	Chain length
meth-	C_1
eth- (acet-)	C_2
prop-	C_3
but-	C_4
pent- (glut-)	C_5
hex-	C_6
hept-	C_7
oct-	C_8
non-	C_9
dec-	C_{10}

Suffix	Functional group
-oic acid	carboxylic acid
-oic anhydride	acid anhydride
-oate	carboxylic ester
-amide	amide
-al	aldehyde
-one	ketone
-ol	alcohol
-amine	amine
-ane	alkane
-ene	alkene

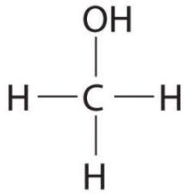
Oxidation states of carbon



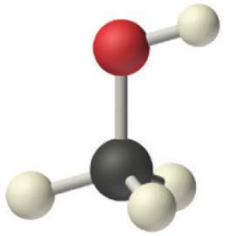
Different ways to represent organic molecules



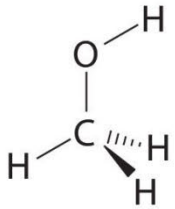
Molecular formula



Structural formula



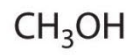
Ball-and-stick model



Perspective drawing

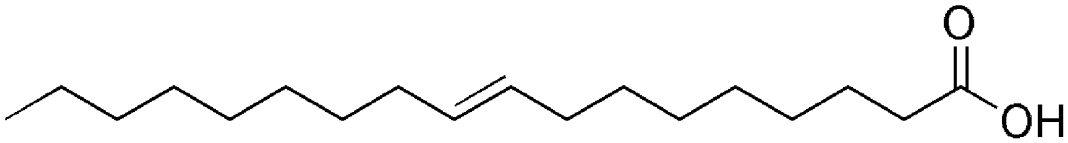


Space filling model

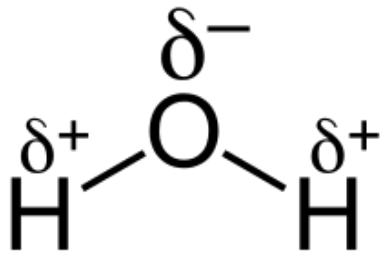


Condensed structural formula

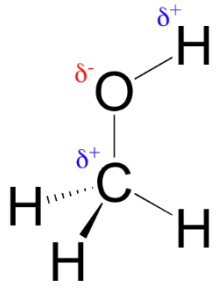
Bond-line formula



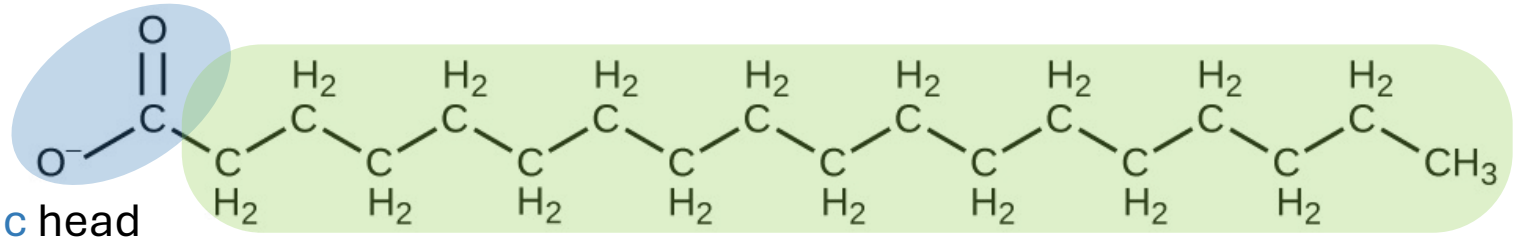
Polarity



Polar → Hydrophilic (soluble in water)



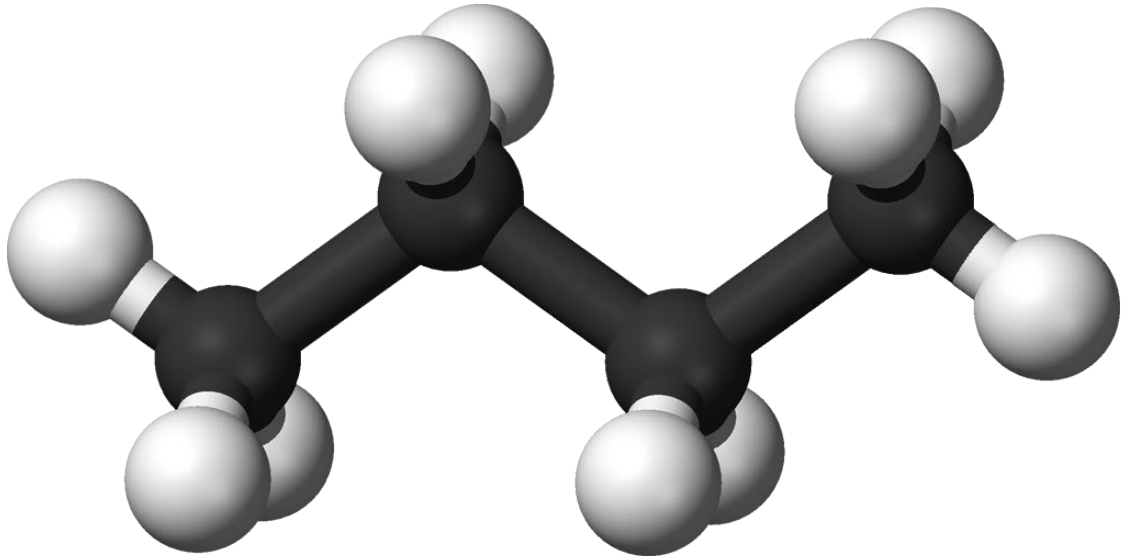
Nonpolar → Hydrophobic (insoluble in water)

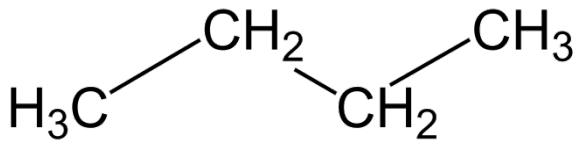


Hydrophilic head

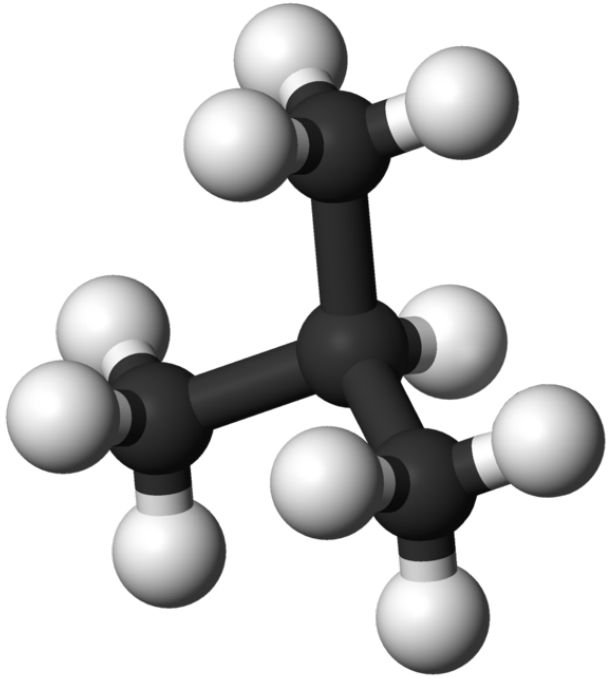
Hydrophobic tail

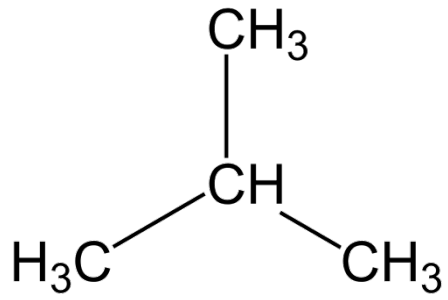
Isomerism





Butane





Isobutane

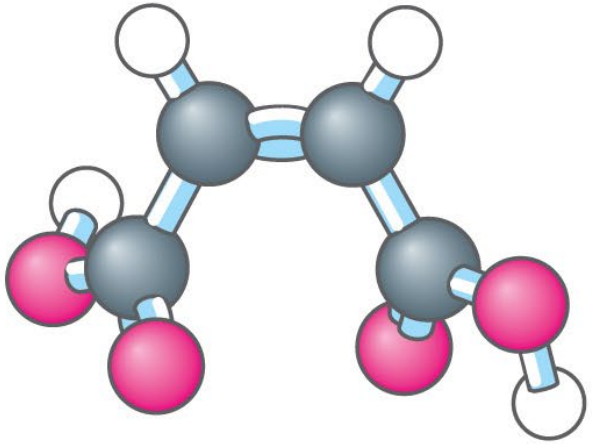
Isomerism

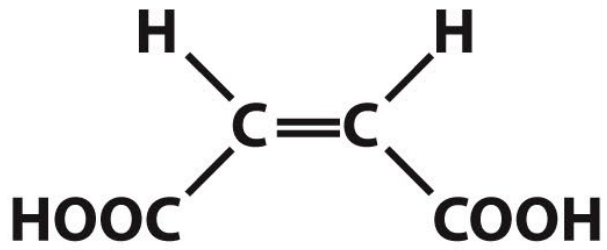
Isomers: compounds that have the same molecular formula but a different arrangement of atoms (which results in different physical/chemical properties)

- Structural isomers
  different ways of connecting the atoms
  - Chain isomers
  - Position isomers
  - Functional isomers
- Stereoisomers
  different 3D orientations of the atoms
  - Geometric isomers
  - Optical isomers

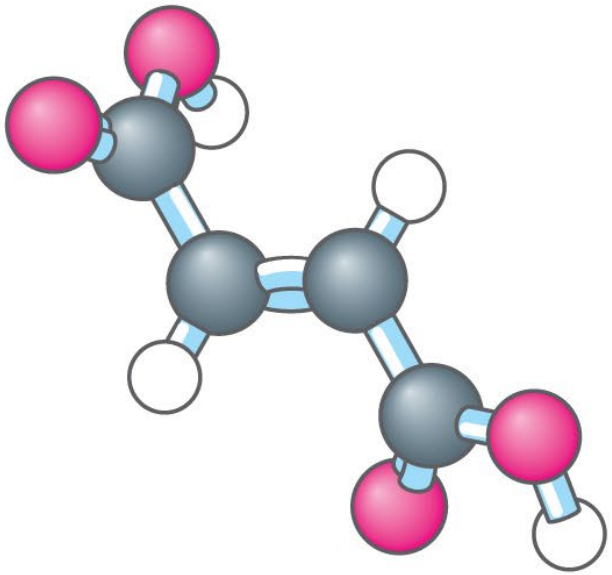
Isomers cannot be interconverted without breaking covalent bonds

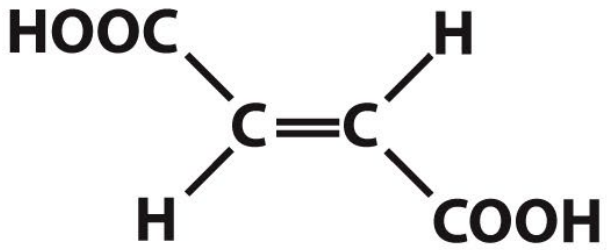
Geometric isomers (cis vs. trans)





Maleic acid (cis)

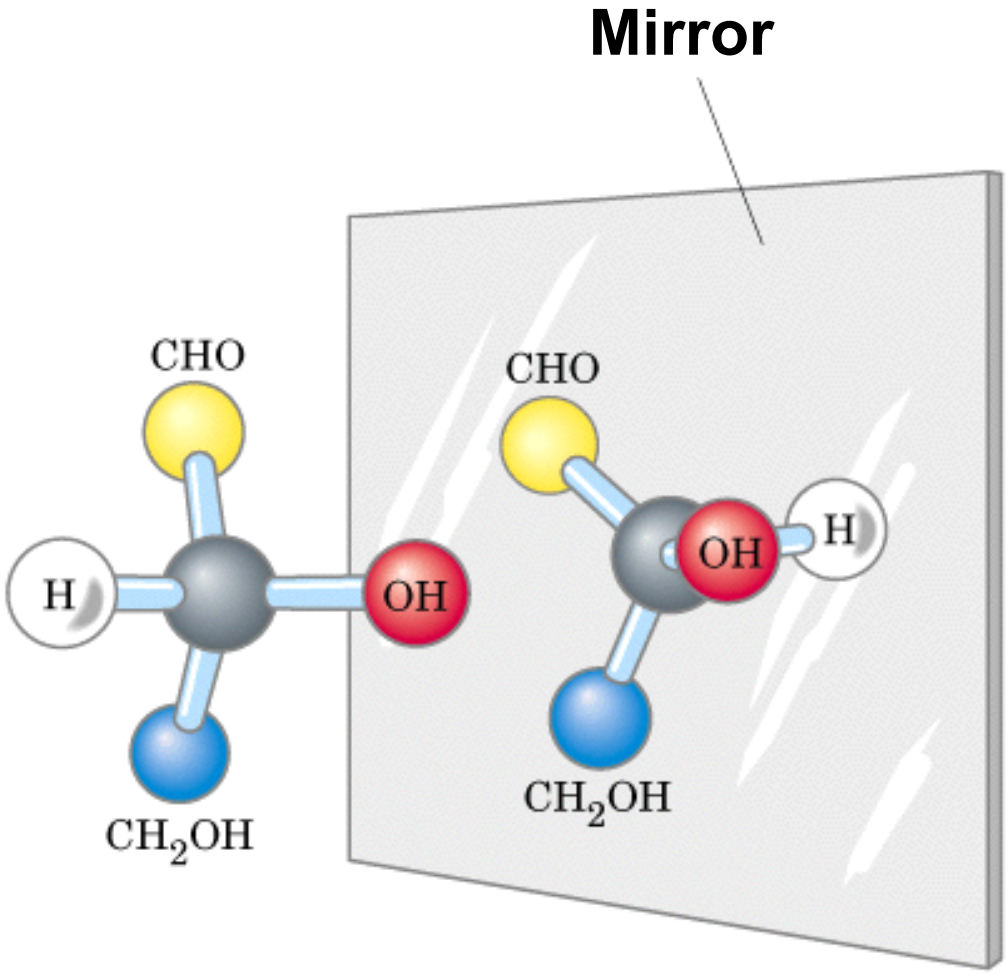




Fumaric acid (trans)

Figure 1-20a
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

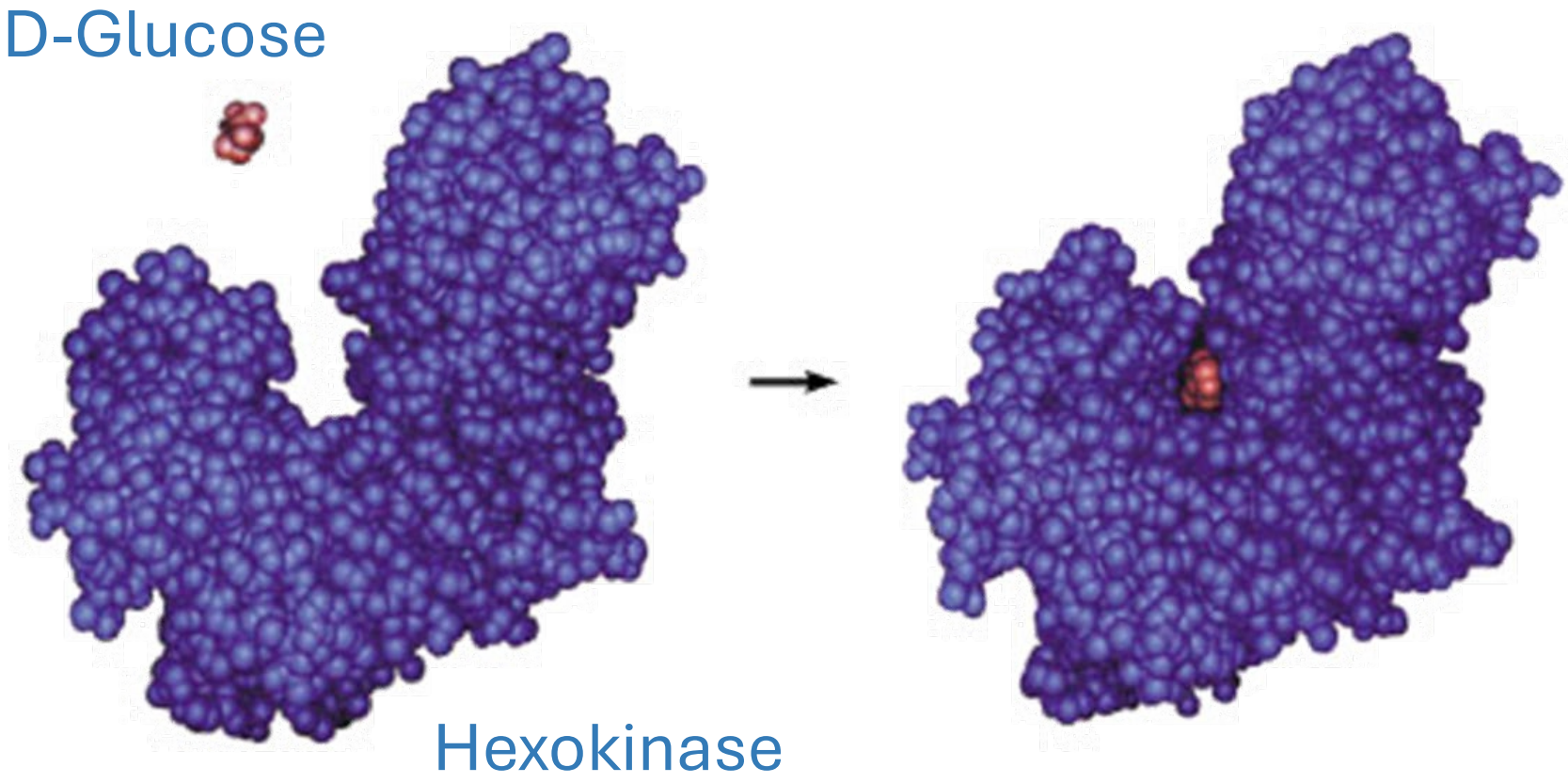
Optical isomers



- Optical isomers have identical chemical and physical properties*
- Biologically, they are completely different

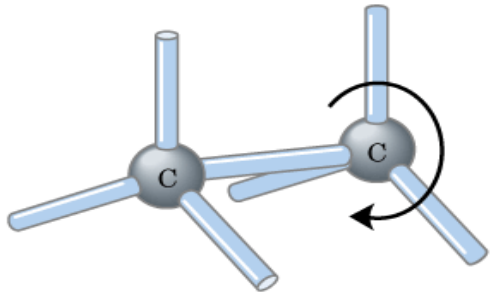
* except that they rotate plane-polarized light in opposite directions

Optical isomers



Configuration vs. conformation

Different conformations of a molecule are obtained by free rotation around single bonds



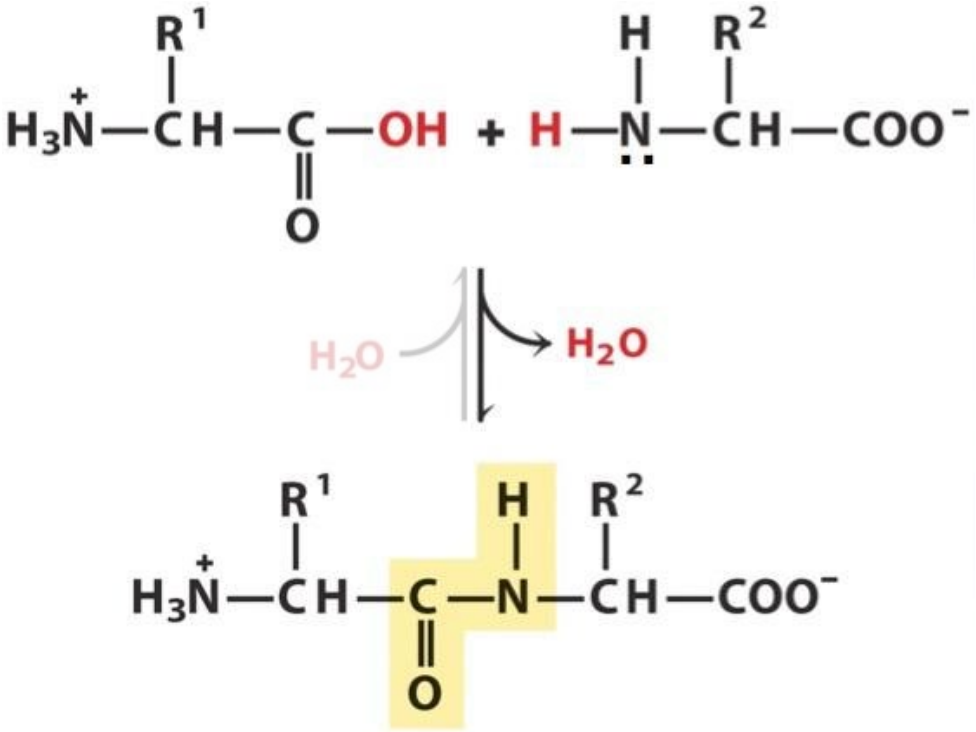
Different configurations cannot be interconverted without breaking bonds (*cis* vs. *trans*, enantiomers…)

Many important biomolecules are polymers

Biomolecule	Monomer	Polymer
carbohydrate	monosaccharide	disaccharide and polysaccharide
protein	amino acid	polypeptide
nucleic acid	nucleotide	DNA and RNA
lipid	fatty acid (and others)	Fat, sphingolipid, etc.

Condensation

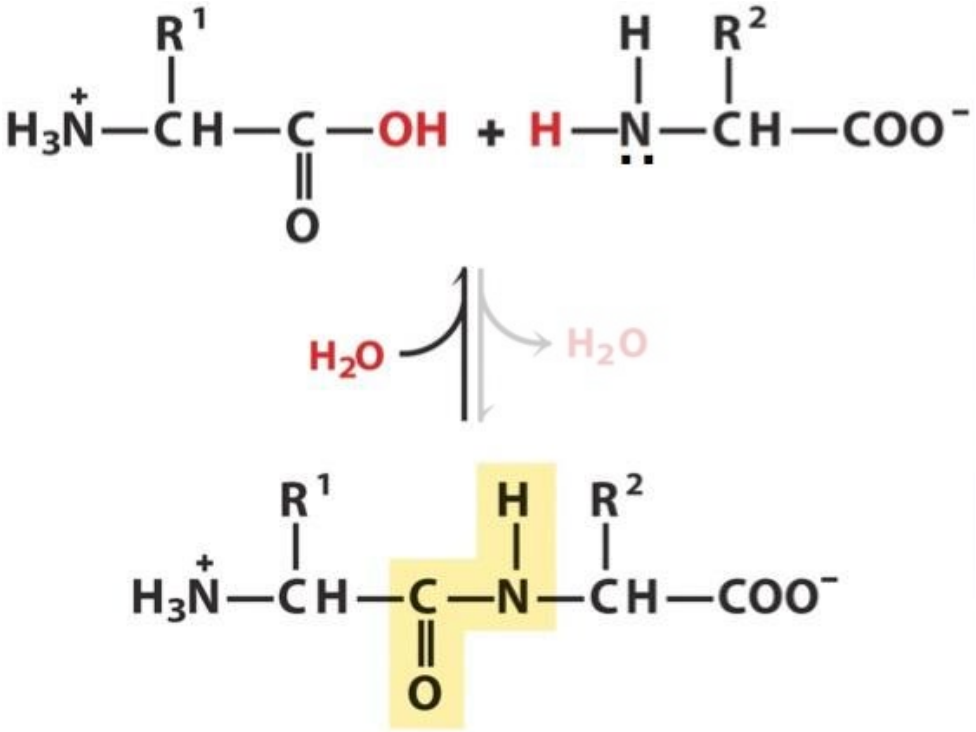
Biopolymers are formed by condensation



Formation of a covalent bond between two molecules by the removal of water

Hydrolysis

Biopolymers are degraded by hydrolysis



Breaking of a covalent bond by reacting with water